

COMPREHENSIVE DESIGN & IMPLEMENTATION REPORT

GENOME ASSEMBLY RESEARCH NETWORK

Course Code: CSE421 (Computer Networks)

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PART 1: NETWORK CONTEXT & VLSM ADDRESSING ARCHITECTURE

1. Introduction & Research Context

Modern bioinformatics labs process vast amounts of raw biological data. In genome assembly pipelines, high-throughput sequencers generate millions of short DNA fragments (reads) that must be dynamically aligned, evaluated, and compiled. This operational pipeline is distributed across six specialized research units, each requiring a dedicated, secure, and logically segmented network domain [1, 2]:

1. **Sequencing Core Lab (SCL) [520 Hosts]:** The primary data generator. Hosts high-throughput sequencing machines that generate raw FASTQ read files. This unit requires the largest address space and hosts the central application servers (DNS, Web, DHCP, Email) [2].
2. **Read Mapping Unit (RMU) [300 Hosts]:** Aligns the raw short reads to reference genomes using highly computational alignment algorithms (such as Bowtie or BWA).
3. **De Bruijn Graph Unit (DBG) [220 Hosts]:** Specialized in *de novo* genome assembly, building graph databases (K-mer overlaps) to assemble genomes without a reference.

4. **Motif Discovery Unit (MDU) [160 Hosts]:** Searches for regulatory patterns and biological motifs within assembled sequences. Hosts its own local DHCP and SMTP servers [2].
5. **Genome Validation Unit (GVU) [90 Hosts]:** Conducts quality control checks on assembled sequences, validating continuity, accuracy, and completeness.
6. **Archive Storage Node (ASN) [45 Hosts]:** The final repository where compiled and verified genome datasets are statically stored [2].

To achieve seamless data exchange, web resource access, and secure inter-unit email communication, this project implements a hierarchical, routed topology using Cisco Packet Tracer [1, 2].

2. Student ID to Base Network Conversion

2.1 Mathematical Mapping

According to the institutional subnetting rules, the base IPv4 network address is dynamically derived from the unique student identifiers (IDs) of the first two group members:

- **Member 1 (MD. Aminul Islam Ornob):** ID 22101569 $\rightarrow$ Extracting the last two digits yields **69**. This value is assigned to the first octet (1^{st} octet) of the IP address.
- **Member 2 (Afaf Binta Shamsuddin):** ID 22101844 $\rightarrow$ Extracting the last two digits yields **44**. This value is assigned to the second octet (2^{nd} octet) of the IP address.
- **Derived Address Block: 69 . 44 . 0 . 0**
- **Assigned Prefix Length: /16** (Subnet Mask: 255 . 255 . 0 . 0)

2.2 Theoretical Analysis: Classful vs. Classless Inter-Domain Routing (CIDR)

In traditional **Classful Addressing**, IP addresses are divided into rigid classes:

- **Class A:** Range 1 . 0 . 0 . 0 to 126 . 0 . 0 . 0 (Default Mask: /8 or 255 . 0 . 0 . 0)
- **Class B:** Range 128 . 0 . 0 . 0 to 191 . 255 . 0 . 0 (Default Mask: /16 or 255 . 255 . 0 . 0)
- **Class C:** Range 192 . 0 . 0 . 0 to 223 . 255 . 255 . 0 (Default Mask: /24 or 255 . 255 . 255 . 0)

Our derived base IP address, 69 . 44 . 0 . 0, technically falls within the Class A range (69 . 0 . 0 . 0 is Class A). If Classful routing rules were applied, our default subnet mask would be /8, which would encompass over 16 million host addresses on a single broadcast domain. This would cause immense broadcast storms and extreme IP address wastage.

By utilizing **Classless Inter-Domain Routing (CIDR)**, we ignore the rigid default class boundaries and assign a custom prefix length of /16 (255 . 255 . 0 . 0) [2]. This gives us a base network block containing $2^{16} = 65,536$ unique IP addresses, allowing us to perform highly efficient variable subnetting down to the individual research units [1, 2].

3. VLSM Subnetting: Sizing Math & Tree Topology

3.1 FLSM vs. VLSM Theory

- **Fixed Length Subnet Masking (FLSM):** Assigns subnets of identical size to all divisions, regardless of their actual host requirements. If FLSM were applied to this project, we would have to size every single subnet to fit our largest division (SCL: 520 hosts), which requires a /22 mask (1024 addresses). This would waste $1024 - 45 = 979$ IP addresses in the ASN unit alone.
- **Variable Length Subnet Masking (VLSM):** Dynamically adjusts the subnet mask of each subnet to match its specific host requirement [2]. This conserves address space and keeps broadcast domains tightly restricted to their necessary boundaries [1, 2].

3.2 The Sorting Constraint (Why We Sort Descending)

When designing a VLSM scheme, **subnets must be allocated in descending order of host requirements (from largest to smallest).**

Larger subnets require larger blocks of contiguous IP addresses. If we allocate smaller subnets (e.g., a /26 for ASN) first, they will fragment the address space. This creates "holes" in the IP block, making it impossible to find a sufficiently large, contiguous block of host bits to accommodate our larger subnets (like the /22 required for SCL) later.

3.3 Subnetting Calculations (Step-by-Step Math)

To calculate the subnet parameters for each unit, we utilize the host-sizing formula: $\text{Usable Hosts} = 2^H - 2$ Where H represents the number of host bits remaining in the subnet mask. The subtractive constant of 2 accounts for the **Network Address** (all host bits set to 0) and the **Direct Broadcast Address** (all host bits set to 1), neither of which can be assigned to an end-device interface.

STEP-BY-STEP SUBNET CALCULATIONS

1. Sequencing Core Lab (SCL) LAN

- **Requirement:** 520 Hosts
- **Calculation:**
 - If we use $H = 9$ host bits: $2^9 - 2 = 510$ usable hosts (Insufficient).
 - If we use $H = 10$ host bits: $2^{10} - 2 = 1022$ usable hosts (Sufficient).
- **Mask Configuration:** $32 - 10 = 22$ network bits. Subnet prefix is **/22** (Subnet Mask: 255.255.252.0).
- **Allocation:**
 - Network Address: **69.44.0.0**

- First Usable Host IP (Default Gateway): **69 . 44 . 0 . 1**
- Last Usable Host IP: **69 . 44 . 3 . 254**
- Broadcast Address: **69 . 44 . 3 . 255**

2. Read Mapping Unit (RMU) LAN

- **Requirement:** 300 Hosts
- **Calculation:**
 - If we use \$H = 8\$ host bits: $2^8 - 2 = 254$ usable hosts (Insufficient).
 - If we use \$H = 9\$ host bits: $2^9 - 2 = 510$ usable hosts (Sufficient).
- **Mask Configuration:** $32 - 9 = 23$ network bits. Subnet prefix is **/23** (Subnet Mask: 255 . 255 . 254 . 0).
- **Allocation:** (Starts immediately at the next available block after 69 . 44 . 3 . 255)
 - Network Address: **69 . 44 . 4 . 0**
 - First Usable Host IP (Default Gateway): **69 . 44 . 4 . 1**
 - Last Usable Host IP: **69 . 44 . 5 . 254**
 - Broadcast Address: **69 . 44 . 5 . 255**

3. De Bruijn Graph Unit (DBG) LAN

- **Requirement:** 220 Hosts
- **Calculation:**
 - If we use \$H = 8\$ host bits: $2^8 - 2 = 254$ usable hosts (Sufficient).
- **Mask Configuration:** $32 - 8 = 24$ network bits. Subnet prefix is **/24** (Subnet Mask: 255 . 255 . 255 . 0).
- **Allocation:**
 - Network Address: **69 . 44 . 6 . 0**
 - First Usable Host IP (Default Gateway): **69 . 44 . 6 . 1**
 - Last Usable Host IP: **69 . 44 . 6 . 254**
 - Broadcast Address: **69 . 44 . 6 . 255**

4. Motif Discovery Unit (MDU) LAN

- **Requirement:** 160 Hosts
- **Calculation:**
 - If we use \$H = 8\$ host bits: $2^8 - 2 = 254$ usable hosts (Sufficient).

- **Mask Configuration:** $32 - 8 = 24$ network bits. Subnet prefix is **/24** (Subnet Mask: 255.255.255.0).
- **Allocation:**
 - Network Address: **69.44.7.0**
 - First Usable Host IP (Default Gateway): **69.44.7.1**
 - Last Usable Host IP: **69.44.7.254**
 - Broadcast Address: **69.44.7.255**

5. Genome Validation Unit (GVU) LAN

- **Requirement:** 90 Hosts
- **Calculation:**
 - If we use $H = 6$ host bits: $2^6 - 2 = 62$ usable hosts (Insufficient).
 - If we use $H = 7$ host bits: $2^7 - 2 = 126$ usable hosts (Sufficient).
- **Mask Configuration:** $32 - 7 = 25$ network bits. Subnet prefix is **/25** (Subnet Mask: 255.255.255.128).
- **Allocation:**
 - Network Address: **69.44.8.0**
 - First Usable Host IP (Default Gateway): **69.44.8.1**
 - Last Usable Host IP: **69.44.8.126**
 - Broadcast Address: **69.44.8.127**

6. Archive Storage Node (ASN) LAN

- **Requirement:** 45 Hosts
- **Calculation:**
 - If we use $H = 6$ host bits: $2^6 - 2 = 62$ usable hosts (Sufficient).
- **Mask Configuration:** $32 - 6 = 26$ network bits. Subnet prefix is **/26** (Subnet Mask: 255.255.255.192).
- **Allocation:**
 - Network Address: **69.44.8.128**
 - First Usable Host IP (Default Gateway): **69.44.8.129**
 - Last Usable Host IP: **69.44.8.190**
 - Broadcast Address: **69.44.8.191**

7. Shared Backbone WAN (RMU-DBG-MDU)

- **Requirement:** 3 Router Interfaces connected to a shared switch.
- **Calculation:**
 - We need at least 3 usable IP addresses.
 - If we use $H = 3$ host bits: $2^3 - 2 = 6$ usable hosts (Sufficient).
- **Mask Configuration:** $32 - 3 = 29$ network bits. Subnet prefix is **/29** (Subnet Mask: 255.255.255.248).
- **Allocation:**
 - Network Address: **69.44.8.192**
 - First Usable IP (RMU): **69.44.8.193**
 - Second Usable IP (DBG): **69.44.8.194**
 - Third Usable IP (MDU): **69.44.8.195**
 - Last Usable IP: **69.44.8.198**
 - Broadcast Address: **69.44.8.199**

8. Point-to-Point Serial WAN Links

- **Requirement:** 2 Usable IPs per point-to-point link.
- **Calculation:**
 - If we use $H = 2$ host bits: $2^2 - 2 = 2$ usable hosts (Sufficient).
- **Mask Configuration:** $32 - 2 = 30$ network bits. Subnet prefix is **/30** (Subnet Mask: 255.255.255.252).
- **Allocations:**
 - **SCL to RMU Link:** Network 69.44.8.200/30 (Usable: .201 - .202, Broadcast: .203)
 - **SCL to DBG Link:** Network 69.44.8.204/30 (Usable: .205 - .206, Broadcast: .207)
 - **SCL to GVV Link:** Network 69.44.8.208/30 (Usable: .209 - .210, Broadcast: .211)
 - **MDU to GVV Link:** Network 69.44.8.212/30 (Usable: .213 - .214, Broadcast: .215)
 - **DBG to MDU Link:** Network 69.44.8.216/30 (Usable: .217 - .218, Broadcast: .219)

- **ASN to GVV Link:** Network 69.44.8.220/30 (Usable: .221 - .222, Broadcast: .223)

3.4 Summary VLSM Subnetting Table

(Copy this table directly to your final documentation)

Unit / Link Name	Required Hosts	Subnet Mask	CIDR	Network Address	Usable IP Range	Broadcast Address
Sequencing Core Lab (SCL)	520	255.255.252.0	/22	69.44.0.0	69.44.0.1 - 69.44.3.254	69.44.3.255
Read Mapping Unit (RMU)	300	255.255.254.0	/23	69.44.4.0	69.44.4.1 - 69.44.5.254	69.44.5.255
De Bruijn Graph Unit (DBG)	220	255.255.255.0	/24	69.44.6.0	69.44.6.1 - 69.44.6.254	69.44.6.255
Motif Discovery Unit (MDU)	160	255.255.255.0	/24	69.44.7.0	69.44.7.1 - 69.44.7.254	69.44.7.255
Genome Validation (GVU)	90	255.255.255.128	/25	69.44.8.0	69.44.8.1 - 69.44.8.126	69.44.8.127
Archive Storage Node (ASN)	45	255.255.255.192	/26	69.44.8.128	69.44.8.129 - 69.44.8.190	69.44.8.191
Backbone Switch (RMU-DBG-MDU)	3	255.255.255.248	/29	69.44.8.192	69.44.8.193 - 69.44.8.198	69.44.8.199
SCL to RMU WAN Link	2	255.255.255.252	/30	69.44.8.200	69.44.8.201 - 69.44.8.202	69.44.8.203
SCL to DBG WAN Link	2	255.255.255.252	/30	69.44.8.204	69.44.8.205 - 69.44.8.206	69.44.8.207

Unit / Link Name	Required Hosts	Subnet Mask	CIDR	Network Address	Usable IP Range	Broadcast Address
					06	
SCL to GVV WAN Link	2	255.255.255.252	/30	69.44.8.208	69.44.8.209 - 69.44.8.210	69.44.8.211
MDU to GVV WAN Link	2	255.255.255.252	/30	69.44.8.212	69.44.8.213 - 69.44.8.214	69.44.8.215
DBG to MDU WAN (Backup)	2	255.255.255.252	/30	69.44.8.216	69.44.8.217 - 69.44.8.218	69.44.8.219
ASN to GVV WAN Link	2	255.255.255.252	/30	69.44.8.220	69.44.8.221 - 69.44.8.222	69.44.8.223

4. High-Yield Viva Preparation Guide (Subnetting & Addressing)

Prepare your team members by having them study these critical questions often asked by faculty during project defense:

- **Q1: Why do we subtract 2 in the host calculation formula ($2^H - 2$)?**
 - *Answer:* The two reserved addresses are the **Network Address** (where all host bits are binary 0) which is used to identify the route in routing tables, and the **Direct Broadcast Address** (where all host bits are binary 1) which is used to send a packet to every host on that local subnet simultaneously. Neither can be assigned to an actual NIC interface.
- **Q2: What is the difference between Classless Inter-Domain Routing (CIDR) and traditional Classful Addressing?**
 - *Answer:* Classful addressing restricts networks to fixed subnet mask boundaries (/8, /16, or /24). CIDR allows for **Classless Inter-Domain Routing**, where network prefixes can be designated at any arbitrary bit boundary (e.g., /22, /25, /30). This prevents massive IP address wastage.
- **Q3: Why must we sort host requirements descendingly when applying VLSM?**
 - *Answer:* Larger subnets require larger contiguous blocks of address spaces. If we subnet smaller networks first, we risk dividing the address space into fragments that are too small to fit the binary boundaries required for the larger subnets later.

- **Q4: Under what IP class does our base network 69 . 44 . 0 . 0 fall?**
 - *Answer:* It falls under **Class A** (range 1 . 0 . 0 . 0 to 127 . 0 . 0 . 0). However, we are subnetting it classlessly using a /16 mask [2].
 - **Q5: What is a Subnet Mask and what is its role?**
 - *Answer:* A 32-bit sequence composed of contiguous binary 1s (representing network bits) followed by contiguous binary 0s (representing host bits). Its role is to help a network interface perform a bitwise AND operation on an IP address to determine if a destination IP is on the **local** broadcast domain or a **remote** network requiring a default gateway.
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PART 2: PHYSICAL LAYER, MODULE SELECTION, AND CABLING INFRASTRUCTURE

1. Hardware Selection & Router Chassis Modification

1.1 Device Classifications

To construct an enterprise-grade genome assembly research network, specific hardware models were selected to match the required bandwidth, processing power, and modularity [2]:

- **Cisco 2911 Integrated Services Router (ISR):** Selected as the Layer 3 routing engine for each research unit. The 2911 offers three onboard 10/100/1000 Mbps RJ-45 GigabitEthernet ports, a highly modular architecture supporting High-Speed WAN Interface Cards (HWICs), and sufficient hardware-based encryption and processing capacity to handle large FASTQ genomic datasets.
- **Cisco Catalyst 2960 Series Switch:** Deployed at Layer 2 for local area networks (LANs). This switch provides 24 FastEthernet ports (up to 100 Mbps) for local end-host connectivity (PCs, Printers, and Servers) and 2 dual-purpose GigabitEthernet uplink ports to establish high-speed connections to the local router's gateway interface.

1.2 Modular Expansion via HWIC-2T

A standard Cisco 2911 router has only three built-in GigabitEthernet interfaces. In this design, central routers like SCL and GVU must connect to multiple neighboring routers and local LANs simultaneously. Because Ethernet ports are limited, point-to-point **Serial connections** were used to establish the WAN backbone [2].

To provide these Serial interfaces, **High-Speed WAN Interface Cards (HWIC-2T)** were physically installed into each router's expansion slots [2].

- **HWIC-2T Features:** Each module provides two serial ports utilizing the Cisco smart serial connector, capable of supporting synchronous bidirectional data transfer rates up to 8 Mbps per port.

- **Redundancy Allocation:** Every router was expanded with **two** HWIC-2T modules [2]. This ensures that each unit has at least four physical serial ports, supporting the primary WAN paths and any backup/floating links.

1.3 Physical Slot Mapping & Port Naming

In Cisco routers, interface names are determined by the slot in which the modular card is physically inserted. The naming convention follows the format: `Interface_Type Slot_Number [Subslot_Number] Port_Number`

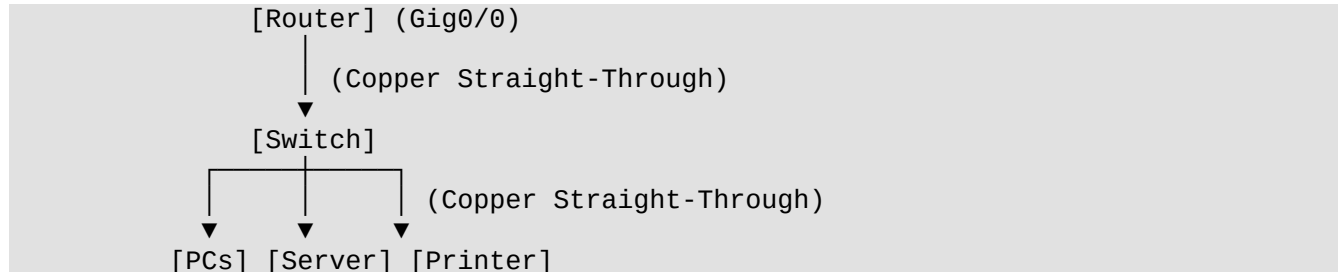
During the physical installation in Packet Tracer, the HWIC-2T modules were placed into **Slot 2** and **Slot 3** of the router chassis [2]:

- **Slot 2:** Automatically mapped ports **Serial0/2/0** and **Serial0/2/1**.
- **Slot 3:** Automatically mapped ports **Serial0/3/0** and **Serial0/3/1**.

This slot mapping was documented to ensure that all CLI configuration scripts matched the actual physical layout, preventing command errors during implementation [2].

2. Cabling Logic and Media Selection

The network utilizes distinct cabling standards to accommodate different transmission distances, speeds, and physical layer requirements:



2.1 Copper Straight-Through Cables (TIA/EIA 568A/568B)

Used for all LAN and Backbone Ethernet connections.

- **Application:** Connecting different device types (e.g., PC NIC to Switch port, Server to Switch port, Switch uplink to Router GigabitEthernet interface) [2].
- **Media Specifications:** Category 6 (Cat6) Unshielded Twisted Pair (UTP) copper cabling terminated with RJ-45 connectors, capable of supporting Gigabit Ethernet speeds (up to 1000 Mbps) with a maximum run distance of 100 meters.

2.2 Serial DCE (Data Communications Equipment) Cables

Used to connect routers over point-to-point WAN links [2].

- **Application:** Router-to-router connections (e.g., SCL to RMU, SCL to DBG, etc.) [2].

- **DCE vs. DTE:** In a Serial WAN link, one side must act as the **DCE (Data Communications Equipment)** and the other as the **DTE (Data Terminal Equipment)**. The DCE side physically dictates the communication speed by providing the clocking signal. In this topology, SCL and GVU host the DCE interfaces and are configured with `clock rate 64000` to synchronize data transfer [2].
- **Clocking Speed:** A clock rate of 64,000 bps was chosen to simulate low-bandwidth, long-distance WAN links, making the dynamic routing and failover scenarios highly observable during testing [2].

3. Detailed Cabling & Interface Mapping Matrix

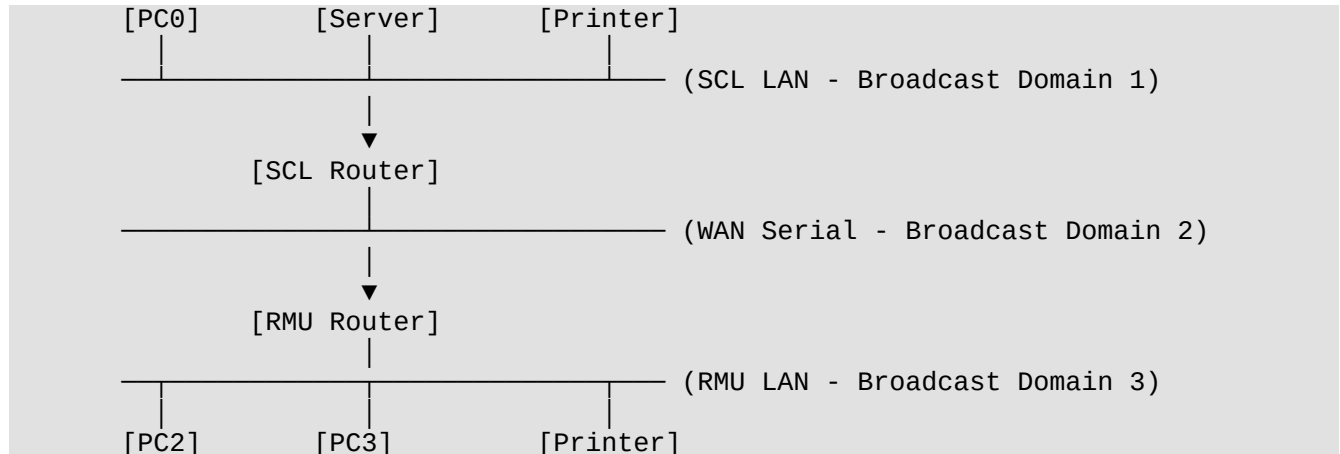
The following matrix provides a complete port-to-port mapping of the physical layout [2]. It serves as the primary reference guide for your team during installation, testing, and troubleshooting:

Source Device	Source Port	Destination Device	Destination Port	Cable Type	Physical Role
SCL Router	Gig0/0	Switch0 (SCL)	Gig0/1	Cat6 Straight-Through	SCL LAN Gateway
SCL Router	Se0/2/0	RMU Router	Se0/2/0	Serial DCE (Clock: 64k)	SCL-RMU Primary WAN [2]
SCL Router	Se0/2/1	DBG Router	Se0/2/0	Serial DCE (Clock: 64k)	SCL-DBG Primary WAN [2]
SCL Router	Se0/3/0	GVU Router	Se0/2/0	Serial DCE (Clock: 64k)	SCL-GVU Primary WAN [2]
RMU Router	Gig0/0	Switch1 (RMU)	Gig0/1	Cat6 Straight-Through	RMU LAN Gateway
RMU Router	Gig0/1	Switch6 (Backbone)	Gig0/1	Cat6 Straight-Through	Backbone Multi-Access [2]
DBG Router	Gig0/0	Switch2 (DBG)	Gig0/1	Cat6 Straight-Through	DBG LAN Gateway
DBG Router	Gig0/1	Switch6 (Backbone)	Gig0/2	Cat6 Straight-Through	Backbone Multi-Access [2]
DBG Router	Se0/2/1	MDU Router	Se0/2/1	Serial DCE (Clock: 64k)	DBG-MDU Backup WAN [2]
MDU Router	Gig0/0	Switch3	Gig0/1	Cat6 Straight-	MDU LAN

Source Device	Source Port	Destination Device	Destination Port	Cable Type	Physical Role
		(MDU)		Through	Gateway
MDU Router	Gig0/1	Switch6 (Backbone)	Fa0/1	Cat6 Straight-Through	Backbone Multi-Access [2]
MDU Router	Se0/2/0	GVU Router	Se0/2/1	Serial DTE (Clock: None)	MDU-GVU WAN Link [2]
GVU Router	Gig0/0	Switch4 (GVU)	Gig0/1	Cat6 Straight-Through	GVU LAN Gateway
GVU Router	Se0/3/0	ASN Router	Se0/2/0	Serial DCE (Clock: 64k)	GVU-ASN WAN Link [2]
ASN Router	Gig0/0	Switch5 (ASN)	Gig0/1	Cat6 Straight-Through	ASN LAN Gateway

4. LAN vs. WAN Segment Isolation

To maintain secure, organized communications between the genomics research units, the network establishes clear boundaries between LAN and WAN domains:



4.1 Broadcast Domains

- **Definition:** A logical division of a computer network in which all nodes can reach each other by broadcast at the Data Link Layer (Layer 2).
- **Isolation Logic:** Switches propagate broadcast frames (such as ARP Requests and DHCP Discover messages) to every single connected port. If our entire network were connected via switches, a single broadcast would flood all six units, consuming valuable bandwidth.
- **The Router's Role:** Routers completely block Layer 2 broadcast frames at their interfaces. By connecting each LAN switch to a unique physical interface on its local router (Gig0/0), we

isolate each research unit into its own **independent broadcast domain**. Broadcast traffic inside SCL cannot cross into RMU or DBG.

4.2 Collision Domains

- **Definition:** A physical segment of a network where data packets can collide with one another during transmission on shared media.
 - **Isolation Logic:** Older network hubs operated on half-duplex, meaning only one device could transmit at a time, creating a single collision domain.
 - **The Switch's Role:** Modern Catalyst 2960 switches operate on **full-duplex**, dedicate an isolated internal path for each port, and maintain a MAC Address Table. Because of this, **every single physical port on a switch is its own individual collision domain**. Collisions are physically impossible in this design, ensuring maximum throughput for raw genomic data transfers.
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5. High-Yield Viva Preparation Guide (Physical & Hardware)

Prepare your team by reviewing these hardware and physical layer questions:

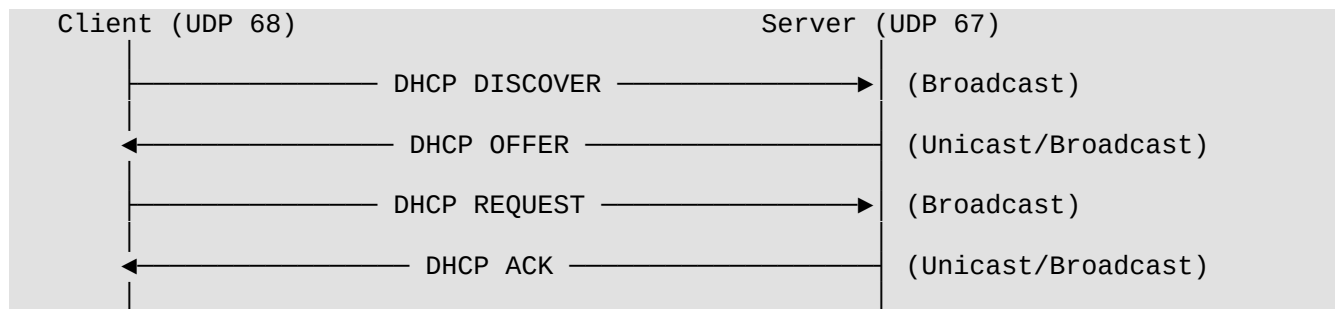
- **Q1: What is the purpose of configuring a clock rate on a Serial interface?**
 - *Answer:* Serial connections require synchronization between transmitting and receiving devices. The side configured with the `clock rate` command acts as the **DCE (Data Communications Equipment)** and provides the physical clocking signal to synchronize the data transmission speed over the link.
- **Q2: What will happen if you connect a Serial cable but forget to configure a clock rate on the DCE side?**
 - *Answer:* The line protocol will remain "**down**" (indicated by red triangles in Packet Tracer), and no data packet exchange will be possible because the devices cannot synchronize their transmission rates.
- **Q3: Why did we choose the Cisco 2911 ISR instead of a standard generic router model?**
 - *Answer:* The Cisco 2911 is a modular enterprise router with high-throughput Gigabit Ethernet ports, supporting multiple HWIC slot expansions. This modularity was required to add the serial HWIC-2T modules while maintaining dedicated Gigabit Ethernet links for the LAN gateways and backbone network [2].
- **Q4: How do switches and routers differ in terms of segmenting Collision and Broadcast domains?**
 - *Answer:* A switch segments **collision domains** (every switch port is an isolated collision domain), but maintains a **single broadcast domain** across all its ports. A router segments **broadcast domains**; every physical router interface is an isolated broadcast domain that blocks Layer 2 broadcasts from passing through.
- **Q5: What is the difference between a modular router and a fixed router?**

- *Answer:* A fixed router has a set, unchangeable number of physical interfaces built into its chassis. A modular router (such as the Cisco 2911 ISR) contains empty expansion slots (HWIC/network module slots), allowing engineers to add custom interfaces (such as Serial or fiber ports) based on the specific needs of the network topology [2].

PART 3: HYBRID DHCP SERVICES & RELAY ARCHITECTURES

1. Theoretical Analysis of the DHCP Protocol

Dynamic Host Configuration Protocol (DHCP) is a client/server protocol that automates the assignment of IP addresses, subnet masks, default gateways, and DNS server addresses to end-hosts [2]. It eliminates the need for manual, error-prone static configuration, conserving administrative overhead.



1.1 The DORA Process (Step-by-Step)

DHCP operates at the Application Layer using **UDP port 67** (for the server) and **UDP port 68** (for the client). The IP leasing process follows the four-step **DORA** transaction:

1. **D - Discover:** The client boots up with no IP address and broadcasts a **DHCP Discover** packet (Source IP: 0.0.0.0, Destination IP: 255.255.255.255, Source MAC: Client NIC MAC, Destination MAC: FF:FF:FF:FF:FF:FF). This packet is flooded to all ports in the local collision and broadcast domain to locate any available DHCP server.
2. **O - Offer:** Upon receiving the broadcast, any active DHCP server reserves an available IP address from its pool and sends a **DHCP Offer** containing the suggested IP, subnet mask, lease duration, default gateway, and DNS server IP.
3. **R - Request:** The client receives the offer. If multiple offers are received, it accepts the first one and broadcasts a **DHCP Request** to formally claim the address. This broadcast informs the chosen server that its offer is accepted and signals other servers to return their offered IPs to their pools.
4. **A - Acknowledge:** The server receives the request and replies with a unicast **DHCP Acknowledgment (ACK)** packet, confirming the lease and finalizing the client's network configuration.

2. Server-Based DHCP Configurations

In **Server-Based DHCP**, dedicated host devices are configured as DHCP servers [2]. This centralizes address management, making it highly scalable and easy to troubleshoot. In this project, dedicated servers are deployed in the **SCL** and **MDU** LANs [2].

2.1 SCL Dedicated DHCP Server (69.44.0.5)

Because SCL hosts the central server farm, its DHCP server is responsible for allocating IP addresses to both its own local LAN (**SCL LAN**) and the remote **Genome Validation Unit (GVU)** LAN [2].

- **SCL-LAN Pool Configuration:**

- *Pool Name:* serverPool
- *Default Gateway:* 69.44.0.1
- *DNS Server:* 69.44.0.2
- *Start IP Address:* **69.44.0.7**
- *Subnet Mask:* 255.255.252.0
- *Why .7?* The first six addresses (69.44.0.1 through 69.44.0.6) are reserved for static devices: the router interface, central servers, and the local printer [2]. Starting the pool at .7 prevents IP duplicate conflicts.

- **GVU-Pool Configuration:**

- *Pool Name:* GVU-Pool
- *Default Gateway:* 69.44.8.1
- *DNS Server:* 69.44.0.2
- *Start IP Address:* **69.44.8.3**
- *Subnet Mask:* 255.255.255.128
- *Why .3?* The first two addresses in the GVU LAN are reserved statically for the gateway (69.44.8.1) and the printer (69.44.8.2) [2].

2.2 MDU Dedicated DHCP Server (69.44.7.3)

The Motif Discovery Unit operates its own local DHCP server to handle its 160 required hosts [2]:

- **MDU-LAN Pool Configuration:**

- *Pool Name:* serverPool
- *Default Gateway:* 69.44.7.1
- *DNS Server:* 69.44.0.2

- *Start IP Address:* **69 . 44 . 7 . 5**
 - *Subnet Mask:* 255 . 255 . 255 . 0
 - *Why .5?* The first four addresses are statically allocated to the local gateway (69 . 44 . 7 . 1), Email server (69 . 44 . 7 . 2), the DHCP server itself (69 . 44 . 7 . 3), and the printer (69 . 44 . 7 . 4) [2].
-

3. Router-Based DHCP Configurations (with host offset exclusions)

In **Router-Based DHCP**, the Cisco IOS router is configured to run the DHCP service directly on its own hardware [2]. This eliminates the need for an external server in smaller units. In this project, **RMU** and **DBG** units utilize router-based pools [2].

3.1 RMU Router-Based DHCP

The RMU DHCP pool must start at a host offset equal to the last two digits of **Member 1's Student ID (69)**:

- **Sizing logic:** RMU's local LAN network is 69 . 44 . 4 . 0/23. To start address leasing exactly at host IP **69 . 44 . 4 . 69**, we must exclude the first 68 host addresses of the subnet.
- **Cisco IOS Configuration:**

```
RMU(config)# ip dhcp excluded-address 69.44.4.1 69.44.4.68
RMU(config)# ip dhcp pool RMU-Pool
RMU(dhcp-config)# network 69.44.4.0 255.255.254.0
RMU(dhcp-config)# default-router 69.44.4.1
RMU(dhcp-config)# dns-server 69.44.0.2
```

3.2 DBG Router-Based DHCP

The DBG DHCP pool must start at a host offset equal to the last two digits of **Member 2's Student ID (44)**:

- **Sizing logic:** DBG's local LAN network is 69 . 44 . 6 . 0/24. To start address leasing exactly at host IP **69 . 44 . 6 . 44**, we must exclude the first 43 host addresses of the subnet.
- **Cisco IOS Configuration:**

```
DBG(config)# ip dhcp excluded-address 69.44.6.1 69.44.6.43
DBG(config)# ip dhcp pool DBG-Pool
DBG(dhcp-config)# network 69.44.6.0 255.255.255.0
DBG(dhcp-config)# default-router 69.44.6.1
DBG(dhcp-config)# dns-server 69.44.0.2
```

4. DHCP Relay & IP Helper Architecture

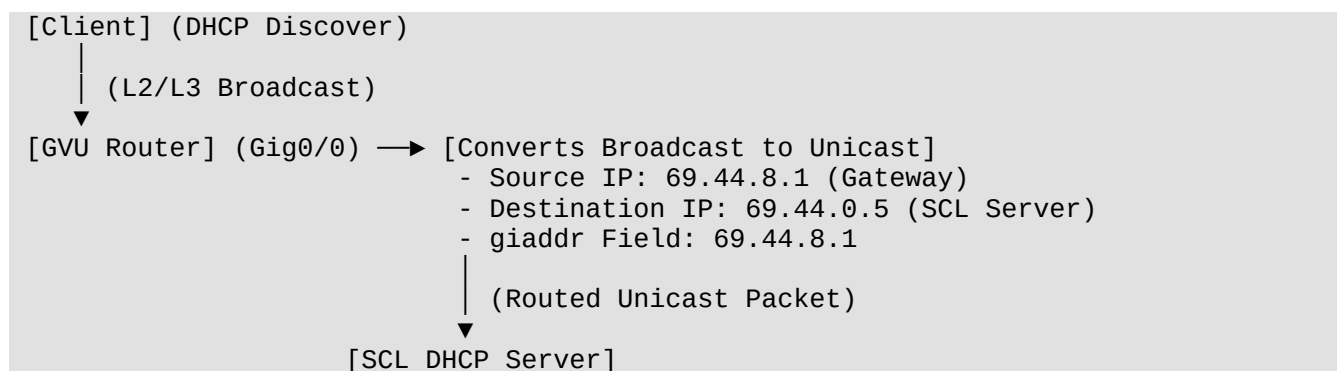
4.1 The Broadcast Boundary Problem

A major limitation of routers is that they **block Layer 2 and Layer 3 broadcasts** to prevent network storms. Because a DHCP Discover message is a broadcast, it cannot cross past the local router interface (Gig0/0).

Consequently, clients in remote LANs (like Gvu) cannot reach a centralized DHCP server (like the SCL DHCP Server at 69.44.0.5) across router boundaries [2].

4.2 The Unicast-to-Broadcast Conversion Solution

To solve this, the **DHCP Relay Agent** (configured via the **ip helper-address** command) is applied on the router interface connected to the client LAN [2].



1. The client in Gvu broadcasts a **DHCP Discover**.
2. The Gvu Router receives this broadcast on interface GigabitEthernet0/0 [2].
3. Because `ip helper-address 69.44.0.5` is configured on Gig0/0, the router intercepts the broadcast, packages the DHCP packet into a **unicast IP packet**, and sets:
 - **Source IP:** 69.44.8.1 (Gvu router interface IP)
 - **Destination IP:** 69.44.0.5 (SCL DHCP Server IP)
 - **giaddr (Gateway IP Address) Field:** 69.44.8.1 (Crucial for subnet matching!)
4. The router forwards this unicast packet across the WAN. SCL DHCP Server receives it.

4.3 How the Server Matches the Correct Subnet (The giaddr Field)

When the central DHCP server receives the unicast request, it reads the **giaddr (Gateway IP Address)** field.

- The server sees `giaddr = 69.44.8.1`.
- It checks its configured pools and finds a pool matching this subnet: the **Gvu-Pool** (network 69.44.8.0/25 with gateway 69.44.8.1) [2].

- It leases an address from the GVV-POOL and sends a unicast reply back to 69.44.8.1. The GVV router then broadcasts the reply locally to the requesting client.

4.4 DHCP Relay CLI Configurations

- **SCL Router Configuration:**

```
SCL(config)# interface GigabitEthernet0/0
SCL(config-if)# ip helper-address 69.44.0.5
```

- **GVV Router Configuration:**

```
GVV(config)# interface GigabitEthernet0/0
GVV(config-if)# ip helper-address 69.44.0.5
```

5. High-Yield Viva Preparation Guide (DHCP Layer)

Prepare your team by practicing these highly technical questions regarding DHCP and Relays:

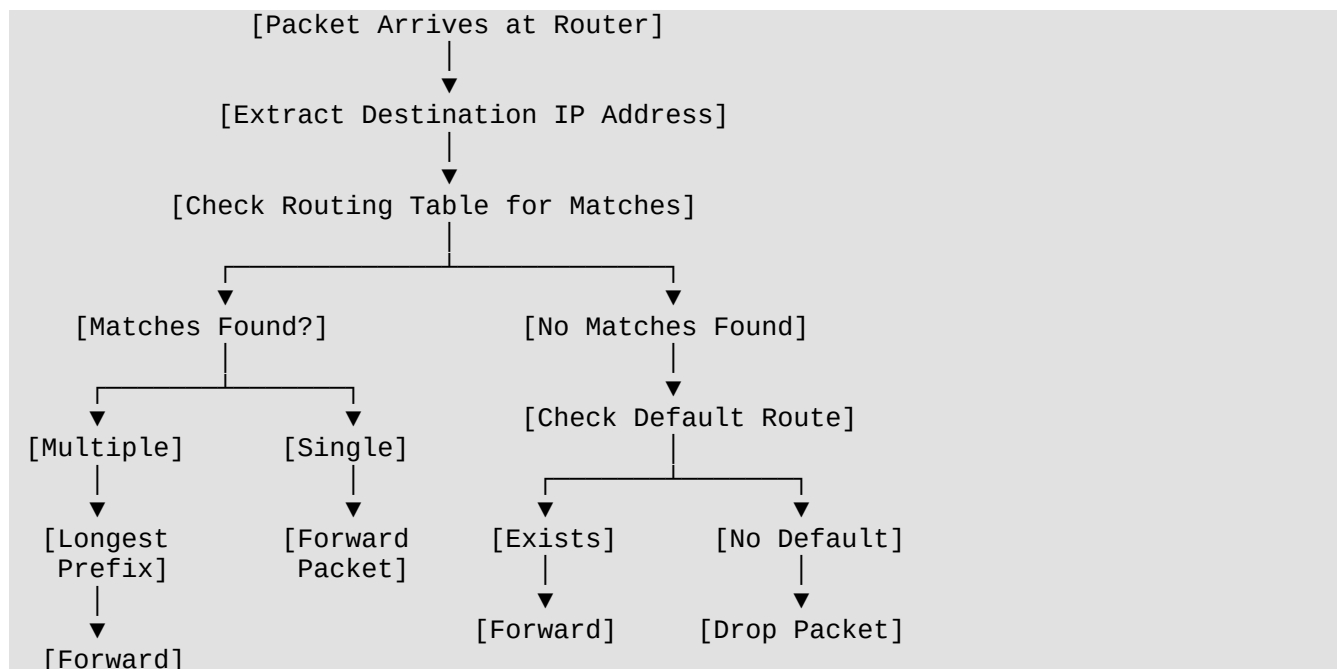
- **Q1: What is APIPA and why does a client get an address like 169.254.x.x?**
 - *Answer:* **Automatic Private IP Addressing (APIPA)** is a link-local address generation mechanism. If a DHCP-configured client sends a DHCP Discover but receives no DHCP Offer within a timeout period, the client automatically assigns itself an address in the reserved Class B range 169.254.0.1 to 169.254.254.254 with a /16 mask, allowing basic communication with other local hosts on that switch [2].
- **Q2: What is the role of the giaddr field in a DHCP Relay packet?**
 - *Answer:* The giaddr (Gateway IP Address) field is populated by the DHCP Relay Agent (the router) with the IP of the interface that received the client's DHCP broadcast. This field tells the centralized DHCP server exactly which subnet pool the client belongs to so the server can assign an address from the correct scope.
- **Q3: Why did PC8 and PC9 (GVV LAN) fail to receive DHCP addresses *before* we configured routing, even though the ip helper-address was active?**
 - *Answer:* The DHCP Relay converts the broadcast into a **routed unicast packet** destined for 69.44.0.5 [2]. Without end-to-end routing (RIPv2 or static), the GVV router did not have a path to route the packet to SCL's LAN, and the SCL router did not have a return route to send the DHCP Offer back to the GVV gateway (69.44.8.1).
- **Q4: Under what protocol and ports does DHCP operate?**
 - *Answer:* DHCP operates under **UDP** at the Transport Layer. Clients use **UDP port 68** to send messages, and DHCP servers use **UDP port 67** to listen and reply.
- **Q5: What is the purpose of the ip dhcp excluded-address command on a Cisco router?**
 - *Answer:* It prevents the router's local DHCP server from leasing out specific IP addresses within its scope. This is essential to safeguard statically configured addresses (such as

default gateways, dedicated servers, and network printers) from being dynamically assigned to PCs, which would cause critical IP address conflicts [2].

PART 4: DYNAMIC & STATIC ROUTING CONFIGURATIONS

1. Fundamental Routing Principles & Path Determination

At the Network Layer (Layer 3), a router's primary function is **path determination** and **packet forwarding** [1, 2]. Routers make decisions by maintaining a **Routing Table**—a database of known network destinations, their associated exit interfaces, and next-hop IP addresses.



1.1 The Longest Prefix Match Rule

When a router receives an IP packet, it reads the destination IP and compares it to all entries in its routing table. If multiple entries match the destination, the router applies the **Longest Prefix Match** rule.

This rule dictates that the router will select the routing entry with the longest prefix length (most specific subnet mask) [2]. For example, if a packet is destined for 69.44.8.135:

- Match A: 69.44.8.0/24 (Prefix length: 24)
- Match B: 69.44.8.128/26 (Prefix length: 26)

The router will select **Match B** because it is a more specific path to that host.

2. Comprehensive Analysis of Configured Static Routes

Static routing is a manual process where the network administrator hardcodes specific routes into a router's table [2]. In this project, standard static routes were avoided to satisfy the rigorous physical constraints, and five distinct routing configurations were applied instead [2]:

2.1 Next-Hop Static Route (SCL)

- **The Constraint:** SCL must be able to reach the remote Archive Storage Node (ASN) LAN (69.44.8.128/26) statically [2].
- **Implementation:** Configured on SCL using GVU's incoming Serial interface IP (69.44.8.210) as the next-hop IP [2].
- **Cisco IOS Syntax:**

```
SCL(config)# ip route 69.44.8.128 255.255.255.192 69.44.8.210
```

- **Why Next-Hop?** By pointing to a next-hop IP instead of a local exit interface, the router does not assume the network is directly attached. This reduces local ARP overhead on the SCL interfaces.

2.2 Exit-Interface & Default Static Route (ASN)

- **The Constraint:** ASN is a stub network with only one way out (through GVU) [2]. It must use exit-interface static routing and a default static route to reach all other networks [2].
- **Implementation:** Configured on ASN as an exit-interface default static route pointing to its local interface Serial0/2/0 [2].
- **Cisco IOS Syntax:**

```
ASN(config)# ip route 0.0.0.0 0.0.0.0 Serial0/2/0
```

- **How it works:** This command matches all packets that do not have a specific entry in ASN's routing table (using wildcard 0.0.0.0 0.0.0.0) and immediately pushes them out of the local Serial0/2/0 interface toward GVU, satisfying both constraints with a single line of code [2].

2.3 Recursive Static Route (RMU)

- **The Constraint:** RMU must be able to reach MDU's LAN (69.44.7.0/24) using recursive static routing [2].
- **Implementation:** Configured on RMU using MDU's Gigabit backbone interface IP (69.44.8.195) as the destination next hop [2].
- **Cisco IOS Syntax:**

```
RMU(config)# ip route 69.44.7.0 255.255.255.0 69.44.8.195
```

- **The "Recursive" Lookup Process:**

1. When RMU gets a packet for MDU LAN (69 . 44 . 7 . 5), it checks the routing table and finds a match pointing to next-hop 69 . 44 . 8 . 195.
2. Because the next hop is not directly connected, RMU must perform a **second lookup** to find how to reach 69 . 44 . 8 . 195 [2].
3. It finds that 69 . 44 . 8 . 195 is accessible via its directly connected backbone interface GigabitEthernet0/1 [2].
4. It successfully forwards the packet. This double-lookup process is known as **Recursive Routing**.

2.4 Floating Static Route (DBG)

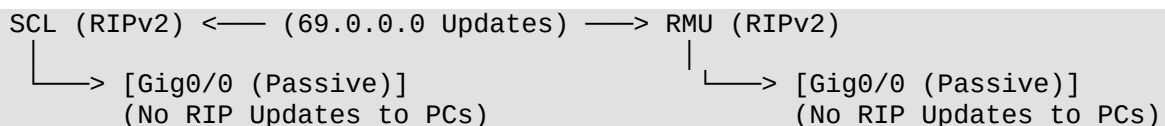
- **The Constraint:** DBG must have an emergency backup path to SCL's LAN (69 . 44 . 0 . 0/22) via MDU, which only activates if the primary dynamic path fails [2].
- **Implementation:** Configured on DBG using MDU's serial interface IP (69 . 44 . 8 . 218) as the next hop, with an **Administrative Distance (AD) of 123** [2].
- **Cisco IOS Syntax:**

```
DBG(config)# ip route 69.44.0.0 255.255.252.0 69.44.8.218 123
```

- **Why AD 123?** Cisco routers select paths based on Administrative Distance (lower is more trusted).
 - Primary Route (RIPv2): AD = **120** [2]
 - Floating Static Route: AD = **123** [2] Because 120 is lower than 123, the router hides the static backup route in its database and uses the RIPv2 path. If the primary link goes down, RIPv2 loses its route, and the router immediately installs this backup route into the active routing table [2].

3. Dynamic Routing Architecture (RIPv2)

To automate routing across the SCL, RMU, DBG, MDU, and GVV routers, **Routing Information Protocol Version 2 (RIPv2)** was deployed [2].



3.1 Why RIPv2?

RIPv2 is a **Distance-Vector** routing protocol that uses a **hop-count metric** (maximum of 15 hops) [2]. Unlike RIPv1, RIPv2 supports **classless routing (CIDR/VLSM)** because it includes subnet mask information inside its routing updates [2]. It also uses multicast address **224 . 0 . 0 . 9** to send updates, reducing overall network processing [2].

3.2 The Classful network Command Constraint

A common point of confusion during project defense is how Cisco IOS processes the RIP network command. When configuring RIP, we type `network 69.0.0.0 [2]`.

Because `69.0.0.0` is a Class A block, Cisco IOS does not support classless subnet masks inside the network command. Even if we typed `network 69.44.0.0`, the IOS would automatically convert it to its classful parent: **network 69.0.0.0 [2]**. This single statement tells the router to activate the RIP process on all physical interfaces on that device that have IPs starting with 69.

3.3 Securing LAN Interfaces via *passive-interface*

By default, once RIPv2 is enabled, it broadcasts routing tables out of all active interfaces, including `GigabitEthernet0/0` (which connects to local switches where end-user PCs reside).

- **The Issue:** Sending routing updates to local PCs wastes LAN bandwidth and poses a major security risk (malicious users on PCs could capture these packets to map your entire network topology).
- **The Solution:** The **passive-interface GigabitEthernet0/0** command was executed on all RIPv2 routers [2]. This stops RIPv2 updates from being sent to client LANs while still allowing the router to advertise that LAN's subnet to other routers.

3.4 GVV Static Redistribution (Making ASN Reachable)

The Archive Storage Node (ASN) cannot run RIPv2 [2]. However, the entire network must still be able to communicate with it.

- **How we solved this:**
 1. We configured a standard static route on GVV pointing directly to ASN's LAN: `ip route 69.44.8.128 255.255.255.192 69.44.8.222 [2]`.
 2. Under the RIPv2 configuration on GVV, we executed the **redistribute static** command [2].
 3. This command tells GVV to take its local static route to ASN and inject it into its RIP dynamic routing updates.
 4. SCL, RMU, DBG, and MDU then automatically learn the route to the ASN LAN via RIPv2 from GVV, satisfying all connectivity requirements [2].

4. Failover Mechanism & Floating Static Route Verification

The floating static route configured on DBG acts as an automated backup system [2]. Here is exactly how the failover behaves under testing:

1. PRIMARY PATH (Active)
DBG Router —[RIPv2 AD 120]—> SCL Router (Primary path is used)
2. FAILOVER EVENT (Link Fails)

DBG Router —[Link Severed ✗]— SCL Router (RIPv2 route is deleted)

3. **BACKUP ACTIVATED (Floating Static)**

DBG Router —[Floating Static AD 123]—> MDU Router —> SCL Router (Backup is used)

Step-by-Step Failover Behavior:

1. **Normal State:** SCL and DBG are directly connected. Both routers run RIPv2. DBG has two paths to SCL LAN (69.44.0.0/22):
 - *Path A (RIPv2):* Directly connected via Serial. AD = **120** [2].
 - *Path B (Static):* Routed through MDU. AD = **123** [2]. Because Path A has a lower AD, the router uses it and inserts it into the routing table (`show ip route` displays Path A with the code **R**).
2. **The Failure Event:** During simulation, SCL's interface `Serial0/2/1` is administratively shut down (simulating a physical link failure).
3. **Route Deletion:** Within seconds, the RIPv2 holddown timer expires. SCL and DBG realize the link is dead, and the primary dynamic route is removed from their routing tables.
4. **Floating Route Activation:** DBG looks at its routing database, sees the floating static route (`ip route 69.44.0.0 255.255.252.0 69.44.8.218 123`), and instantly promotes it to the active routing table [2].
5. **Traffic Rerouted:** Packets from DBG LAN to SCL LAN are now successfully forwarded through the MDU router, maintaining network connectivity [2].
6. **Recovery:** Once SCL's Serial interface is turned back on, RIPv2 re-establishes its connection. DBG sees the lower AD (120) path again, removes the static route from the active table, and restores the primary link [2].

5. High-Yield Viva Preparation Guide (Routing Layer)

Ensure your team is fully prepared for routing-specific questions with this guide:

- **Q1: What is Administrative Distance (AD) and what are its standard values?**
 - *Answer:* AD is a measure of trustworthiness of a routing source (lower is more trusted).
 - Directly Connected: **0**
 - Static Route: **1**
 - RIPv2: **120** [2]
- **Q2: What is a "Floating" Static Route?**
 - *Answer:* A static route configured with a manually altered Administrative Distance that is **higher** than the AD of the active dynamic routing protocol (e.g., AD of 123 vs. RIPv2's

120) [2]. This ensures the static route remains dormant and only installs itself into the routing table if the primary dynamic path fails [2].

- **Q3: How does RIPv2 prevent routing loops?**

- *Answer:* RIPv2 uses several loop-prevention mechanisms:
 - **Split Horizon:** Prevents a router from advertising a route back out of the same interface on which it originally learned it.
 - **Route Poisoning:** When a link fails, the router advertises that route's metric as **16 (infinite/unreachable)**, instantly invalidating it across all neighboring routers.

- **Q4: What does the `no auto-summary` command do in RIPv2?**

- *Answer:* By default, RIP automatically summarizes classless subnets to their classful boundaries (e.g., summarizing 69.44.4.0/23 and 69.44.6.0/24 to Class A 69.0.0.0/8) when sending updates across different networks. The `no auto-summary` command disables this, forcing RIPv2 to advertise the exact, classless subnet masks, which is required for VLSM networks to function properly [2].

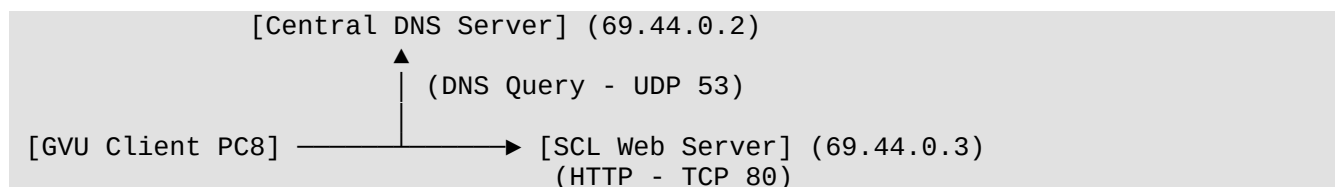
- **Q5: What is the purpose of the `redistribute static` command under router `rip` on the GVV router?**

- *Answer:* It tells the RIPv2 routing process to take any manually configured static routes on GVV (specifically, the static route pointing to the ASN LAN) and advertise them inside RIPv2 updates [2]. This allows SCL, RMU, DBG, and MDU to learn the route to the non-RIP ASN LAN dynamically [2].

PART 5: APPLICATION LAYER SERVICES & SYSTEM VERIFICATION

1. Application Layer Services Configuration Details

The Application Layer is the top-most layer of the OSI model, providing network services directly to end-user applications [1, 2]. In this network, services are centralized in SCL and MDU to support collaborative bioinformatics workflows [2].



1.1 Web Server Configuration (SCL LAN - 69.44.0.3)

The SCL Web Server hosts the institutional bioinformatics portal, allowing researchers across all subnets to retrieve genome assembly reports [2].

- **Protocol & Ports:** **HTTP** (TCP port 80) and **HTTPS** (TCP port 443).
- **Resource Mapping:** The standard `index.html` was customized in the services panel:

```
<html>
<head><title>Genome Assembly Research Network</title></head>
<body style="background-color:#f4f6f9; margin:40px;">
<center>
<h1 style="color:#1a365d; font-family:sans-serif; font-size:36px; margin-
bottom:10px;">
  Genome Assembly Research Network Online
</h1>
<hr style="border: 2px solid #1a365d; width:80%;">
<p style="font-family:sans-serif; font-size:18px; color:#4a5568;">
  Welcome to the Bioinformatics Central Web Services Portal.
</p>
<p style="font-family:sans-serif; font-size:14px; color:#718096;">
  System Status: Operational | Routed Backbone: RIPv2 & Custom Statics
</p>
</center>
</body>
</html>
```

1.2 Central DNS Server Configuration (SCL LAN - 69.44.0.2)

To eliminate the need for researchers to memorize IP addresses, a central Domain Name System (DNS) server is configured [2].

- **Protocol & Ports:** DNS (UDP port 53 for standard queries; TCP port 53 for zone transfers).
- **A-Record Mapping:** The DNS database was populated with three distinct Host (A) records to resolve Fully Qualified Domain Names (FQDNs) to their respective IP addresses [2]:

Hostname (FQDN)	Record Type	Resolved IP Address	Associated Network Service
www.genome-lab.bio	A	69.44.0.3	Bioinformatics Web Portal [2]
mail.sequence.bio	A	69.44.0.4	SCL Unit Mail Exchanger [2]
mail.motif.bio	A	69.44.7.2	MDU Unit Mail Exchanger [2]

1.3 Email Servers Configuration (SMTP & POP3)

Cross-domain institutional emailing allows the SCL and MDU research teams to share DNA alignment analyses and motifs [2]. This configuration implements two independent email domains [2]:

- **SCL Email Server (69.44.0.4):**
 - *Domain Name:* **mail.sequence.bio** [2]
 - *Active Mailbox:* scluser (Password: password)

- *Protocols:* **SMTP** (TCP Port 25 for sending) & **POP3** (TCP Port 110 for receiving).
 - **MDU Email Server (69.44.7.2):**
 - *Domain Name:* **mail.motif.bio** [2]
 - *Active Mailbox:* mduser (Password: password)
 - *Protocols:* **SMTP** (TCP Port 25 for sending) & **POP3** (TCP Port 110 for receiving).
-

2. Step-by-Step Setup of Client Mail Profiles

To enable communication, email profiles are configured on SCL PC0 and MDU PC6 [2]:

2.1 SCL User Profile (PC0)

- Click **PC0** → **Desktop** → **Email** → **Configure Mail**.
- Configure the parameters:
 - *Your Name:* SCL Researcher
 - *Email Address:* scluser@mail.sequence.bio
 - *Incoming Mail Server:* mail.sequence.bio (Resolved by DNS to 69.44.0.4) [2]
 - *Outgoing Mail Server (SMTP):* mail.sequence.bio [2]
 - *Username:* scluser
 - *Password:* password
- Click **Save**.

2.2 MDU User Profile (PC6)

- Click **PC6** → **Desktop** → **Email** → **Configure Mail**.
 - Configure the parameters:
 - *Your Name:* MDU Researcher
 - *Email Address:* mduser@mail.motif.bio
 - *Incoming Mail Server:* mail.motif.bio (Resolved by DNS to 69.44.7.2) [2]
 - *Outgoing Mail Server (SMTP):* mail.motif.bio [2]
 - *Username:* mduser
 - *Password:* password
 - Click **Save**.
-

3. Step-by-Step System Verification Log

The system's end-to-end functionality was verified under standard operating conditions, proving that all configuration phases were successful [2].

3.1 DNS and HTTP Integration Verification

To verify the DNS-to-HTTP mapping, PC8 (GVU LAN) was selected as a remote host.

1. Open the Web Browser on **PC8** [2].
2. Type **www.genome-lab.bio** in the URL box and click **Go** [2].
3. **Result:** The browser successfully queries the DNS server (69.44.0.2), resolves the domain to 69.44.0.3, and renders the customized "Genome Assembly Research Network Online" homepage.

3.2 Cross-Domain Email Transmission Verification (SCL to MDU)

To test cross-domain emailing, an email was composed on PC0 (scluser@mail.sequence.bio) and sent to mduuser@mail.motif.bio [2].

1. PC0 queries DNS (69.44.0.2) to resolve mail.sequence.bio (SMTP)
2. PC0 establishes SMTP (TCP Port 25) session with SCL Email Server (69.44.0.4)
3. SCL Email Server queries DNS to resolve mail.motif.bio (SMTP next hop)
4. SCL Email Server routes mail across WAN to MDU Email Server (69.44.7.2) via SMTP
5. PC6 connects to MDU Email Server via POP3 (TCP Port 110) and pulls email

Step-by-Step Mail Packet Flow:

1. **Local SMTP Upload:** PC0 sends a DNS query to the DNS server (69.44.0.2) to resolve mail.sequence.bio. Once resolved to 69.44.0.4, PC0 establishes an SMTP session (TCP port 25) with the SCL Email Server and uploads the message [2].
2. **Domain Resolution:** SCL Email Server reads the recipient domain (@mail.motif.bio). It queries the DNS server to resolve mail.motif.bio to its destination IP: 69.44.7.2 [2].
3. **Inter-Router Routed Path:** SCL Email Server establishes an SMTP session with the MDU Email Server (69.44.7.2) across the routed WAN. The packets are forwarded by SCL $\rightarrow$ RMU $\rightarrow$ Switch6 (Backbone) $\rightarrow$ MDU.
4. **Local POP3 Retrieval:** The MDU Email Server stores the message in the mduuser mailbox. When the researcher on **PC6** clicks **Receive** on their email app, PC6 establishes a POP3 session (TCP port 110) with the MDU Email server and successfully downloads the email [2].

4. High-Yield Viva Preparation Guide (Application Layer & Verification)

Review these common questions with your team to prepare for your project defense:

- **Q1: What are the differences between SMTP and POP3 in terms of operation and ports?**

- *Answer:* **SMTP (Simple Mail Transfer Protocol)** is a "push" protocol used to upload emails from clients to servers, or to transfer emails between servers. It operates on **TCP port 25** [2]. **POP3 (Post Office Protocol Version 3)** is a "pull" protocol used by clients to download emails from their server mailboxes. It operates on **TCP port 110** [2].
 - **Q2: How does a client determine which DNS server to query?**
 - *Answer:* The client's DNS server IP is assigned during configuration (statically or dynamically via DHCP). When a user enters an FQDN like `www.genome-lab.bio`, the client immediately sends a UDP port 53 query to that configured IP (in this case, `69.44.0.2`) [2].
 - **Q3: What are the main DNS record types?**
 - *Answer:*
 - **A Record (Address):** Maps a hostname to an IPv4 address [2].
 - **AAAA Record:** Maps a hostname to an IPv6 address.
 - **MX Record (Mail Exchanger):** Identifies the mail server for a domain.
 - **CNAME Record (Canonical Name):** Maps an alias hostname to a true hostname.
 - **Q4: Why does DNS run over UDP instead of TCP?**
 - *Answer:* DNS primarily uses **UDP** because DNS queries are small, single-packet requests and responses. UDP is connectionless and has low overhead, allowing for faster domain resolution. TCP is only used if the query response exceeds 512 bytes or during zone transfers between primary and secondary DNS servers.
 - **Q5: What is an FQDN?**
 - *Answer:* A **Fully Qualified Domain Name (FQDN)** is the absolute domain name for a host on the internet, specifying its exact location in the DNS tree hierarchy (e.g., `www.genome-lab.bio.`). It consists of the host name, second-level domain, and top-level domain.
-